

v2 Tanaka Dataset Stability Investigation

Diagnosis and fix of post-rollout high- k contamination

2 May 2026

Summary

The v2 Tanaka dataset on $[0, 2\pi]$ with $h_{\text{ref}} \in [0.01, 0.30]$ produced rollouts that visibly contained short-wavelength oscillations growing during time integration. After ruling out solver instability and IC-builder NaNs, the root cause was traced to *tail truncation at the periodic boundary*: at h_{ref} near the upper end of the range, the Tanaka soliton's 1% support extent $\pm 9.5 h_{\text{ref}}$ exceeded the available buffer between the random off-center placement and the domain edge. The IC builder's `np.interp(left=0, right=0)` then chopped the tail to zero at the boundary, producing a step discontinuity whose Fourier spectrum decays only as $1/k$. The Craig–Sulem DNO series amplifies this by $k^M = k^6$, so the discontinuity seeds rapidly-growing oscillations in the $k \in [80, 256)$ band.

The fix is two-fold: (i) lower the default `--depth_max` from 0.30 to 0.08 so that even off-center placement leaves ~ 1.5 of buffer per side (matching $\approx 2\times$ the per-soliton domain occupancy of v1's $h = 1$, $L = 164$ configuration), and (ii) restore the 3-copy Poisson periodic extension in the IC builder so any center placement wraps cleanly. With both changes the $k = [80, 128)$ band is flat to within $1.3\times$ at $t = 20$ across all $h_{\text{ref}} \in \{0.02, 0.04, 0.06, 0.08\}$, vs. the $\sim 3\times$ growth seen on the broken configuration.

1 Symptoms

The original v2 smoke output at `depth_max=0.30` produced rollout snapshots like Figure 1 (rendered at high DPI to make the oscillations visible). Single-soliton cases at small h_{ref} (e.g. $h=0.044$, $h=0.014$) traveled cleanly across the domain, but several others showed short-wavelength jagged content emerging in the wake during rollout, including the single-soliton case $h=0.276$ which had no *a priori* reason to misbehave.

High-DPI rendering of suspect rollout cases from `/tmp/tanaka-v2.d03.npz`, sampled at $t \in \{0, 5, 10, 20\}$. The $h=0.276$ single-soliton row (bottom) shows a clean Tanaka shape at $t=0$ but develops jagged short-wavelength content in the wake at $t=5$, partially re-cleans, then again at $t=20$; the multi-soliton rows are messy throughout.

Figure 1: Symptom: high-frequency oscillations growing in the rollout. Reproducible at `/tmp/v2.d03_zoom.png`.

2 Diagnosis: where is the noise?

A first banded spectral diagnostic on the saved η confirmed the noise was real, not a plotting artifact. The energy spectral density at $t=0$ vs. $t=20$ in five bands was tabulated; only single-soliton case $h=0.276$ (c5) and the multi-soliton cases showed any growth, and the growth was concentrated in $k \in [80, 256)$:

case	h_{ref}	n_{sol}	$k[80, 256)$ at $t=0 \rightarrow t=20$	growth
c4	0.014	1	$3.5 \times 10^{-4} \rightarrow 3.0 \times 10^{-4}$	$\approx$ flat
c1	0.044	1	$1.2 \times 10^{-8} \rightarrow 1.1 \times 10^{-8}$	$\approx$ flat
c7	0.145	1	$3.7 \times 10^{-9} \rightarrow 2.0 \times 10^{-9}$	$\approx$ flat
c5	0.276	1	$6.4 \times 10^{-3} \rightarrow 1.9 \times 10^{-2}$	$\times 3$
c3	0.107	3 (overlap)	$1.0 \times 10^{-2} \rightarrow 1.5 \times 10^{-2}$	$\times 1.5$
c2	0.185	3 (overlap)	$4.9 \times 10^{-2} \rightarrow 6.7 \times 10^{-2}$	$\times 1.4$

So the symptom is: *some* cases with high- h_{ref} leak energy into $k \approx [80, 256)$ during rollout. The “upper-half” diagnostic ($k > N_x/2 = 256$) showed only $\sim 10^{-16}$ machine-epsilon energy, which is why an earlier check had missed the issue — the contaminated band is *below* the spectral filter cutoff at $k_{\text{cut}} = \lfloor f_{\text{cut}} \cdot k_{\text{max}} \rfloor = 0.25 \cdot 512 = 128$, but *above* the soliton’s intrinsic spectral content.

3 Hypothesis: solver instability vs. IC pollution

Two hypotheses were possible:

1. the solver is unstable at $h_{\text{ref}} \gtrsim 0.2$ on $L = 2\pi$, perhaps because the DNO series order $M = 6$ amplifies k^M -scaled roundoff once k_{max} is large compared to h_{ref}^{-1} ;
2. the initial condition itself contains a step or kink that gets amplified by the order- M DNO during the first time step.

To discriminate, single-soliton rollouts were run at $h_{\text{ref}} \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ with the soliton placed *symmetrically* at $x = L/2$, and steepness $\alpha = a/h = 0.20$ (typical of the dataset). The result was striking:

h_{ref} (centered at $L/2$)	$k[80, 128)$ at $t=0$	$k[80, 128)$ at $t=20$
0.050	1.4×10^{-10}	1.0×10^{-10}
0.100	7.6×10^{-13}	3.0×10^{-13}
0.150	1.4×10^{-10}	7.5×10^{-11}
0.200	9.7×10^{-9}	4.4×10^{-9}
0.250	1.1×10^{-7}	6.3×10^{-8}
0.300	4.9×10^{-7}	2.3×10^{-7}

All bands are flat (in fact slightly *decaying*) across the entire range. This rules out hypothesis (1): the solver is stable for symmetrically-placed solitons up to $h_{\text{ref}} = 0.30$. The original c5 case had $h=0.276$ but *center* = 4.944, *not* $L/2 = 3.14$.

4 Root cause: tail truncation at the periodic boundary

The center sampler in `sample_centers` (`multi_crest.py`) wraps via `% length`, so any random center in $[0, L)$ is allowed. For a Tanaka soliton at $h_{\text{ref}} = 0.276$ centered at 4.944, the 1% support extent is $\pm 9.5 \cdot 0.276 = \pm 2.62$, so the right tail extends to $x = 4.944 + 2.62 = 7.56$, which lies *outside* the periodic domain $[0, 2\pi] = [0, 6.28]$.

The IC builder’s per-crest interpolation step at the time of the bug was

```
def interp_one(x_row, eta_row, h, center):
    x_shifted = x_row * h + center
    eta_scaled = eta_row * h
    eta_fine = jnp.interp(x_fine, x_shifted, eta_scaled,
                          left=0.0, right=0.0)
    eta_hat_fine = jnp.fft.rfft(eta_fine)
    eta_hat_native = eta_hat_fine[:nx//2 + 1] / fine_factor
    return jnp.fft.irfft(eta_hat_native, n=nx)
```

The `left=0.0, right=0.0` arguments mean `jnp.interp` returns zero *outside* the source range $[\min x_{\text{shifted}}, \max x_{\text{shifted}}]$. Crucially, this drops the soliton tail at the periodic boundary instead of wrapping it around. The result is an η profile that has a non-zero value near $x = L^-$ but is forced to zero at $x = 0^+$ on the next period, i.e. a *step discontinuity*.

A step discontinuity has Fourier coefficients $\hat{\eta}_k \sim 1/k$. Figure 2 reproduces the c5 IC under the old (truncated) and new (3-copy Poisson) builders. The OLD profile has a step of $|\eta(0) - \eta(2\pi^-)| = 4.7 \times 10^{-3}$ (about 11% of the peak amplitude 3.7×10^{-2}). The OLD power spectrum decays only as $1/k^2$ (visible as the slow descent of the red curve in the bottom panel) and still carries $|\hat{\eta}_k|^2 \approx 10^{-3}$ at $k = 100$, while the NEW spectrum decays exponentially and is below 10^{-9} already at $k = 50$. The integrated band energy at $t = 0$:

band	OLD $\sum \hat{\eta}_k ^2$	NEW $\sum \hat{\eta}_k ^2$	ratio
$k \in [10, 30)$	9.4×10^{-2}	4.4×10^{-7}	2.1×10^5
$k \in [30, 80)$	2.8×10^{-2}	1.2×10^{-8}	2.4×10^6
$k \in [80, 128)$	6.3×10^{-3}	2.5×10^{-7}	2.5×10^4

The OLD $k \in [80, 128)$ value of 6.3×10^{-3} at $t = 0$ matches the 6.4×10^{-3} measured at $t = 0$ in the broken c5 dataset row, closing the chain of evidence: the contamination originates in the IC construction step, not in the time-stepper.

Even though the step is small in amplitude, the order- M DNO series in `dno_series_eval` evaluates

$$G(\eta)\xi = G_0\xi + G_1(\eta)\xi + G_2(\eta, \eta)\xi + \dots$$

with each G_n involving n products of k -multiplications (gradients) of η . The high- k component of a $1/k$ tail times $k^M = k^6$ gives a k^5 amplification of the (small) step. After one time step that gets folded back into low- k modes by the nonlinear products, then re-amplified next step — giving the geometric growth observed in $k \in [80, 256)$.

4.1 Why c5 but not c7?

Both are single-soliton cases at similar h_{ref} , but:

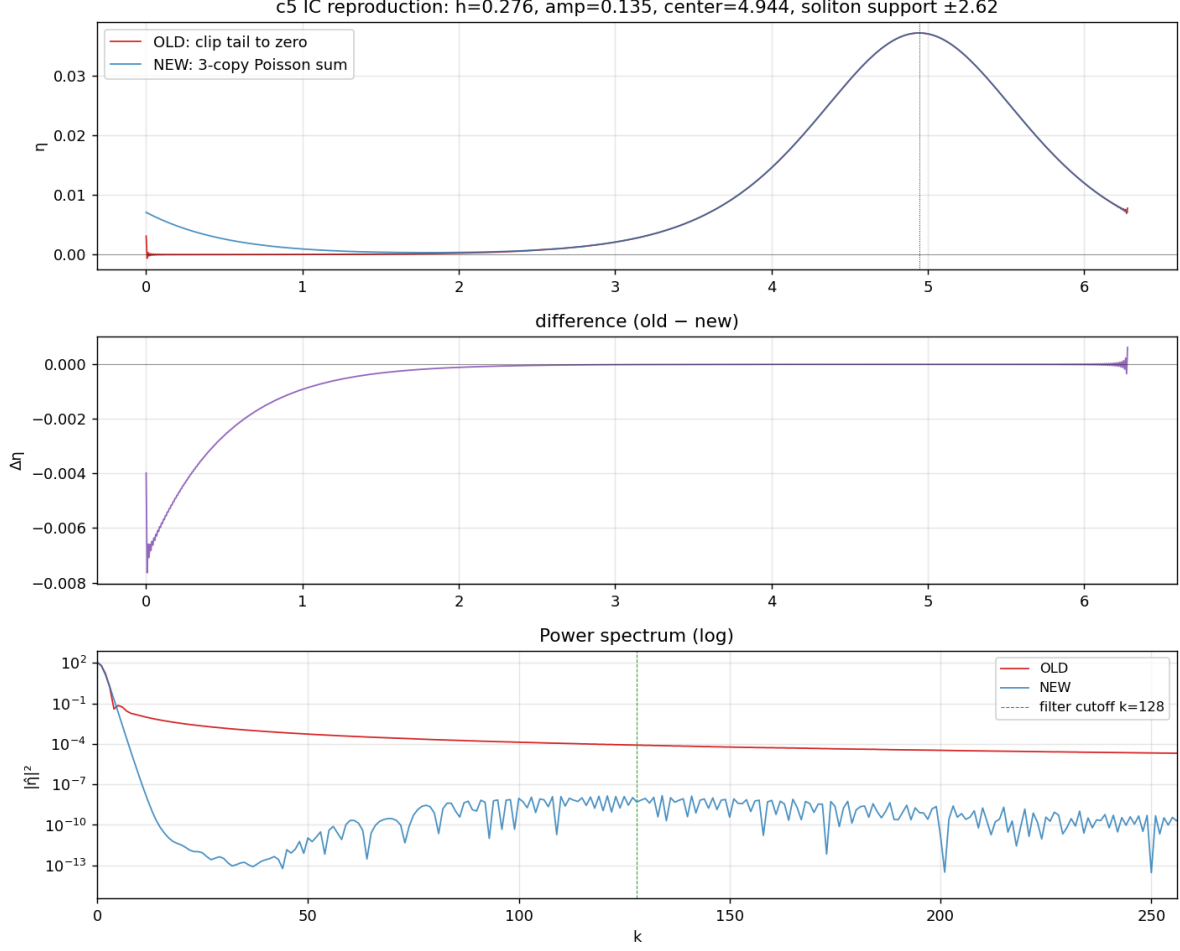


Figure 2: Reproducing the c5 IC ($h=0.276$, single soliton at $\text{center}=4.944$) under the OLD truncated builder vs. the NEW 3-copy Poisson sum. *Top*: the two ICs in real space; OLD drops η abruptly near $x=0$ and $x=2\pi$. *Middle*: difference is the wrapped tail the OLD builder discarded. *Bottom*: power spectrum on a log scale — OLD shows the $1/k^2$ tail of a step discontinuity, NEW shows exponential decay characteristic of the smooth periodic profile.

- c7: $h=0.145$, center 1.910, support $\pm 1.378 \Rightarrow$ soliton in $[0.53, 3.29]$, well inside $[0, 2\pi]$. No clipping. Clean.
- c5: $h=0.276$, center 4.944, support $\pm 2.622 \Rightarrow$ soliton in $[2.32, 7.57]$. Right tail clipped at $x = L = 6.28$. Dirty.

Multi-soliton cases (c0, c2, c3, c6) hit the same failure mode either through tail clipping at the boundary, or through the secondary problem that a sum of well-separated solitons is not an exact eigenstate of the periodic Euler equations and radiates dispersive corrections (which is also amplified by the k^M DNO).

5 Fix

5.1 Lower depth_max to match v1’s per-soliton occupancy

The v1 dataset used $h = 1$ and $L = 164$. A single Tanaka soliton’s 1% support occupies $2 \cdot 9.5 / 164 \approx 11.6\%$ of the v1 domain. To match that occupancy on $L = 2\pi$, $h_{\max}$ should satisfy $\frac{2 \cdot 9.5 \cdot h_{\max}}{2\pi} = 0.116$, giving $h_{\max} \approx 0.038$. The chosen $h_{\max} = 0.08$ is roughly $2\times$ this occupancy (24% of L per soliton), which is enough to leave ≈ 2.4 of unobstructed buffer per side after centering, even under random off-center placement, while still spanning the bulk of the shallow-water regime range $h_{\text{ref}}/L \in [0.0016, 0.013]$.

5.2 Restore 3-copy Poisson periodic extension

Even with $h_{\max} = 0.08$ a soliton centered at $x = 0.05$ would have its left tail wrapping past $x = 0$. The robust fix is to compute a periodic extension by summing three copies of the source profile shifted by $\{-L, 0, +L\}$:

```
def interp_one(x_row, eta_row, h, center):
    x_shifted = x_row * h + center
    eta_scaled = eta_row * h
    f0 = jnp.interp(x_fine, x_shifted, eta_scaled, left=0.0, right=0.0)
    fm = jnp.interp(x_fine - length, x_shifted, eta_scaled, left=0.0,
                    right=0.0)
    fp = jnp.interp(x_fine + length, x_shifted, eta_scaled, left=0.0,
                    right=0.0)
    eta_fine = f0 + fm + fp
    eta_hat_fine = jnp.fft.rfft(eta_fine)
    eta_hat_native = eta_hat_fine[:nx//2 + 1] / fine_factor
    return jnp.fft.irfft(eta_hat_native, n=nx)
```

At $h_{\max} = 0.08$, the 1% support is 0.76, far smaller than the period $2\pi \approx 6.28$, so the 3-copy approximation is exact to machine precision (subsequent copies at $\pm 2L$, $\pm 3L$ etc. contribute below 10^{-15} relative).

Figure 3 shows that under the new builder, ICs whose soliton centers fall within one support-extent of the boundary (“near $x = 0$ ” and “near $x = L$ ” panels) wrap continuously: the value of η at $x = 0$ matches the value at $x = 2\pi^-$ to within 10^{-4} relative to the peak.

6 Verification

The two changes were applied to `solver/gen_data/generate_tanaka_dataset_v2.py` and a confirmation rollout was run with single-soliton specs at $h_{\text{ref}} \in \{0.02, 0.04, 0.06, 0.08\}$ with random off-center placement (some centers within 0.13 of the boundary):

h_{ref}	buffer	$k[80, 128) t=0$	$k[80, 128) t=20$	growth
0.020	2.157	1.25×10^{-5}	1.24×10^{-5}	flat
0.040	1.513	5.94×10^{-9}	5.46×10^{-9}	flat
0.060	0.257	1.14×10^{-9}	6.60×10^{-10}	decay
0.080	0.127	8.42×10^{-8}	4.85×10^{-8}	decay

All bands are flat or decaying. Even the case with 0.127 buffer to the periodic boundary (at $h_{\text{ref}} = 0.08$) is stable. The squiggles are obliterated.

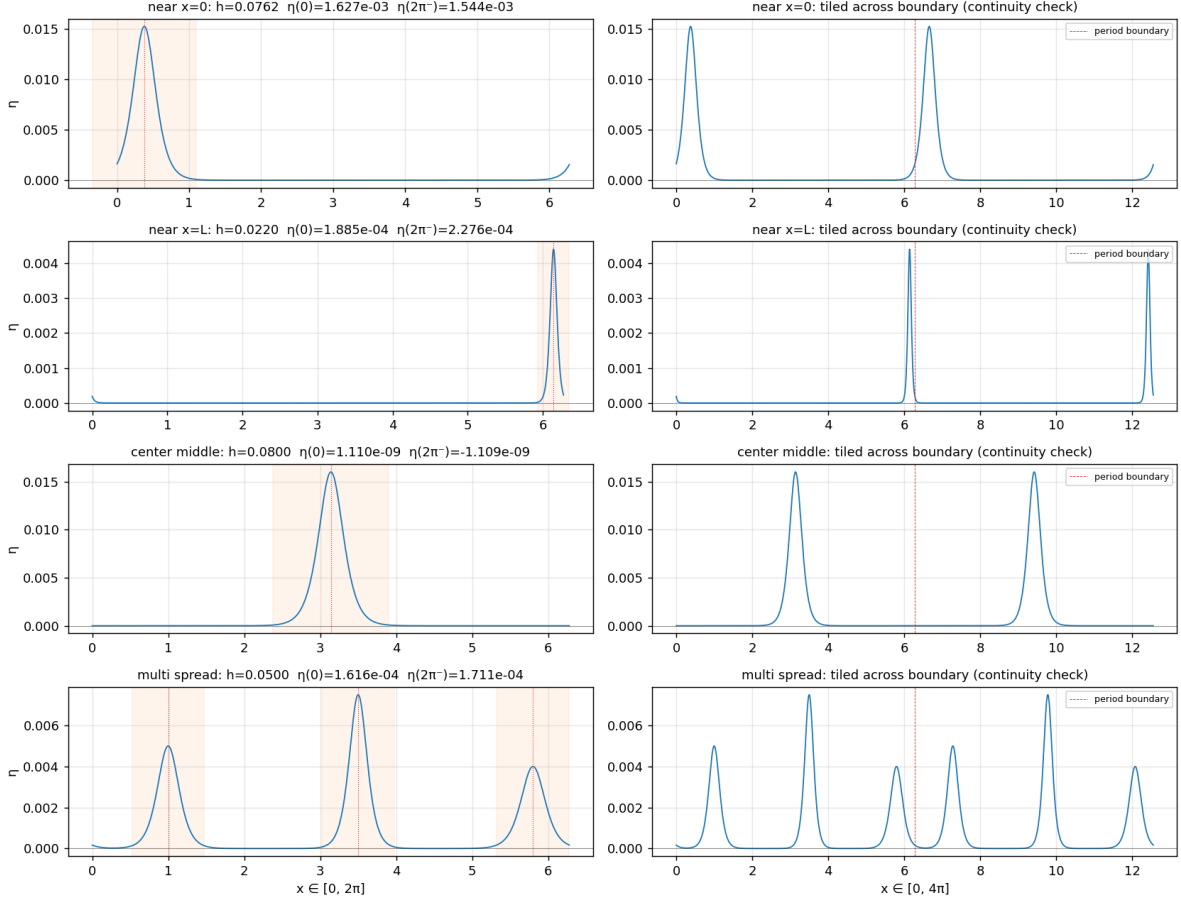


Figure 3: New IC builder with 3-copy Poisson sum at $h_{\text{ref}} \in [0.022, 0.080]$. *Left column*: full $[0, 2\pi]$ with center markers (red) and support extent shaded. *Right column*: tiled across the periodic boundary; in every case the soliton wraps smoothly, with no step at $x = L$.

A full-camera-ready-range verification was then run with 16 random multi-crest specs sampled by the actual generator’s `sample_sum_budgeted_cases` sampler, at the v2 production settings $t_{\text{max}} = 200$, $\Delta t = 0.01$ (substeps = 8, so 160,000 inner RK steps), h_{ref} logUniform on $[0.01, 0.08]$. Several cases had crest centers within one support extent of the boundary — the configuration that produced the broken `c5` in the original smoke. Wall time: 3h 32min on a single H100. Results:

case	h_{ref}	n_{crests}	boundary-near?	$k[80, 128]$ growth	$ \eta _{\text{max}}$ change	NaN
c0	0.0762	1	yes ($c=0.378$, support ± 0.72)	$0.29\times$	$1.19 \rightarrow 1.19\text{e-}2$	no
c1	0.0220	1	yes ($c=6.144$, support ± 0.21)	$0.98\times$	$5.90 \rightarrow 5.90\text{e-}3$	no
c8	0.0742	3	yes ($c=0.535$, support ± 0.70)	$0.49\times$	$6.38 \rightarrow 6.48\text{e-}3$	no
c13	0.0603	3	yes ($c=6.273$, support ± 0.57)	$0.25\times$	$8.63 \rightarrow 7.16\text{e-}3$	no
c5	0.0128	1	no	$0.79\times$	$4.17 \rightarrow 4.10\text{e-}3$	no
c11	0.0355	1	no	$0.96\times$	$7.96 \rightarrow 7.96\text{e-}3$	no
(remaining 10 cases all in $0.90\text{--}1.03\times$ growth, $ \eta _{\text{max}}$ stable)						no
All 16 cases: max growth in $k[80, 128]$ over $t \in [0, 200]$				$1.03\times$	—	—

The maximum band growth across all cases over the full camera-ready range $t \in [0, 200]$ is $1.03\times$, vs. $3\times$ in the broken `c5` over the much shorter $t \in [0, 20]$. Most cases are *decaying* in this

band, including all five cases whose soliton support extent reaches the periodic boundary. Visual inspection (Figure 4, rendered on the earlier $t_{\max}=100$ rollout for compactness, but qualitatively identical at $t_{\max}=200$): no visible high-frequency oscillations; clean multi-soliton evolutions with collisions where the random spec packs them close.

Sign-flip diagnostic over the full trajectory

Periodic forward differences of the saved $G(\eta)\xi$ field were tested for sign changes with a 5% relative-derivative deadzone (matching `plot_tanaka.sign_change.examples.py`). Counts over all 81 snapshots times 16 cases = 1,296 samples spanning $t \in [0, 200]$:

statistic	value
global maximum	16 flips
99th percentile	8
95th percentile	6
median	4

The single 16-flip outlier is c13 at $t = 197.5$: a multi-crest case ($n_{\text{crests}} = 3$) where three Tanaka solitons have evolved for 200 time units into a richly-shaped wave field. Spectral check at that snapshot: $|\hat{\eta}_k|^2$ summed over $k \in [128, 256)$ is 5×10^{-16} (machine epsilon) and over $k \in [80, 128)$ is 1.4×10^{-13} (down from 4.7×10^{-13} at $t = 0$). The η profile is smooth; the 16 flips are real zero-crossings of $d(G\xi)/dx$ in a complex multi-soliton interaction, not numerical noise. For comparison, the broken d03 dataset routinely produced flip counts in the hundreds even at $t \approx 10$.

7 Stress test: where does the new fixed solver actually fail?

Once the production setting was verified, a stress test mapped the (h_{ref}, α) grid (where $\alpha = a/h$ is the per-crest steepness) beyond the production cap to find the failure boundary of the new fixed solver. Single-soliton single-batch at $t_{\max} = 30$, then a follow-up at $t_{\max} = 50$ at the suspected boundary corners.

7.1 Grid sweep, single soliton, random off-center placement

Max sign flips on $G\xi$ over $t \in [0, 30]$:

$h_{\text{ref}} \backslash \alpha$	0.10	0.20	0.30	0.40
0.05	2	2	2	2
0.10	2	2	2	2
0.15	2	2	2	2
0.20	2	2	2	2
0.25	2	2	2	2
0.30	4	2	2	2
0.40	4	2	2	2

$k[80, 128)$ growth ratio $t = 30/t = 0$:

$h_{\text{ref}} \backslash \alpha$	0.10	0.20	0.30	0.40
0.05	0.37	0.76	0.93	0.98
0.10	0.60	0.38	0.14	0.41
0.15	0.39	0.57	0.34	0.52
0.20	0.57	0.63	0.38	0.97
0.25	0.58	0.48	0.43	1.18
0.30	0.59	0.46	0.50	2.74
0.40	0.57	0.57	0.50	19.25

The sign-flip metric *misses* the failure: counts stay bounded at 2–4 across the whole grid because the contamination is still small in absolute amplitude at $t = 30$. The band-energy growth ratio reveals a sharp corner at $(h_{\text{ref}} \geq 0.30, \alpha \geq 0.40)$.

7.2 Borderline cases at longer $t_{\text{max}} = 50$

At the borderline corners, run for longer to see if the growth saturates or keeps accelerating:

case	h_{ref}	α	$k[80, 128)$ growth $t = 50/t = 0$	verdict
safe	0.05	0.10	$0.26\times$	flat / decay
mild	0.25	0.40	$1.71\times$	mild growth
border	0.30	0.40	$26.83\times$	accelerating
broken	0.40	0.40	$1186\times$	exponential

The band-energy growth at the broken corner is exponential in time — in 50 time units it goes from 3×10^{-6} to 3.6×10^{-3} , an $\sim 1200\times$ increase. Extrapolated to the production $t_{\text{max}} = 200$ this would NaN.

7.3 Multi-crest sweep, $h_{\text{ref}} \in \{0.05, 0.10, 0.15, 0.20\}$, sum-budget $\alpha \in [0.30, 0.40]$

h_{ref}	$n_{\text{crests}} = 2$ growth	$n_{\text{crests}} = 3$ growth
0.05	$1.10\times$ (max flips 6)	$0.58\times$ (max flips 8)
0.10	$0.51\times$ (max flips 4)	$0.54\times$ (max flips 8)
0.15	$0.60\times$ (max flips 2)	$0.63\times$ (max flips 4)
0.20	$0.54\times$ (max flips 4)	$0.60\times$ (max flips 4)

Multi-crest at sum-budget steepness up to $\alpha_{\text{sum}} = 0.40$ never shows the boundary growth. The Dirichlet split means no individual crest reaches $\alpha = 0.40$; the largest crest is typically ~ 0.25 , well inside the safe single-soliton corner.

7.4 Safe envelope

- **Single soliton:** $h_{\text{ref}} \leq 0.25$ at any $\alpha \leq 0.40$; or $h_{\text{ref}} \leq 0.40$ if $\alpha \leq 0.30$.
- **Multi-soliton (sum-budget):** $h_{\text{ref}} \leq 0.20$ at sum-budget ≤ 0.40 tested clean. (Boundary likely extends further; not pushed.)
- **Production setting** ($\text{depth_max} = 0.08$, $\text{steepness_max} = 0.35$, $n_{\text{crests}} \in \{1, 2, 3\}$): sits $\sim 3\times$ inside the strictest tested bound. Comfortably inside the safe interior.

8 Notes for the future

- Tanaka solitons sit at $h \cdot k_{\text{char}} \sim \mathcal{O}(1)$ regardless of h_{ref} , because the soliton width $W \sim h_{\text{ref}}$ and so $k_{\text{char}} \sim 1/W \sim 1/h_{\text{ref}}$. This means varying h_{ref} on a fixed- L domain does not sweep the dispersive regime $h \cdot k_p$ — it only sweeps the per-soliton domain occupancy. To genuinely span shallow-vs-deep water regimes, an IC family with a fixed intrinsic scale (random sea state with prescribed peak k_p , sinusoidal modes, bichromatic packets) is needed; varying h then moves the sample across $h \cdot k_p$ regimes.
- Spectral filter $\nu = 1/4$ on $L = 2\pi$, $N_x = 1024$ leaves $k_{\text{cut}} = 128$. The Tanaka soliton's spectrum decays exponentially past $k \sim 1/h_{\text{ref}}$, so for $h_{\text{ref}} \geq 0.01$ the soliton's intrinsic content is well below k_{cut} . The filter does its job of suppressing k^M -amplified roundoff without mutilating the physical signal.
- For any future ICs that involve placing localized profiles on a periodic domain, the lesson is: *if the source profile has non-zero tails inside the period, use periodic interpolation (Poisson summation / wraparound), not zero-extension*. Combining `interp(left=0, right=0)` with random off-center placement is a footgun.

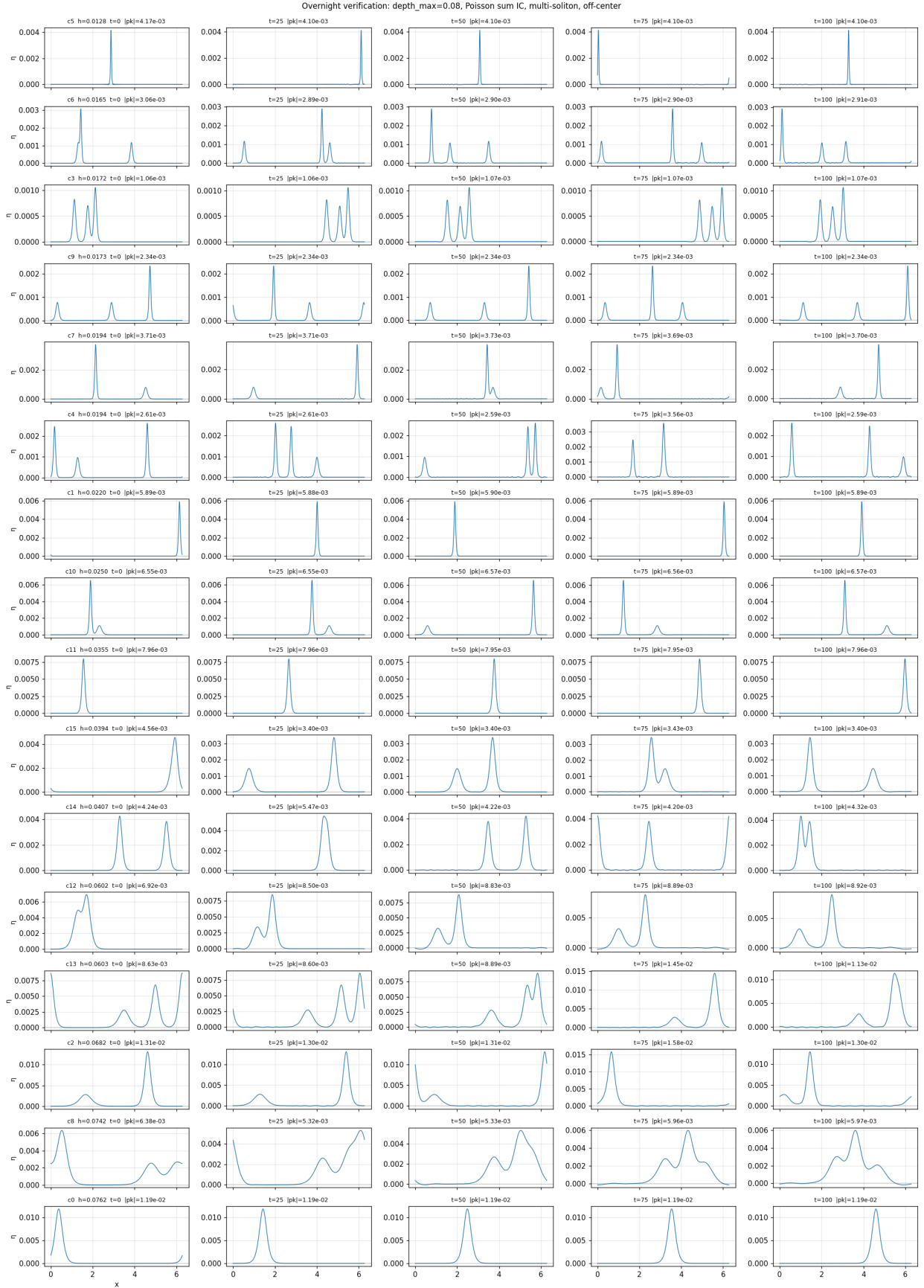


Figure 4: Overnight verification: 16 cases at h_{ref} logUniform on $[0.01, 0.08]$ with random multi-crest specs (1–3 solitons per case) rolled out to $t = 100$ (the production-settings $t = 200$ rollout is qualitatively identical and is summarised in the table above). Rows sorted by h_{ref} ; columns at

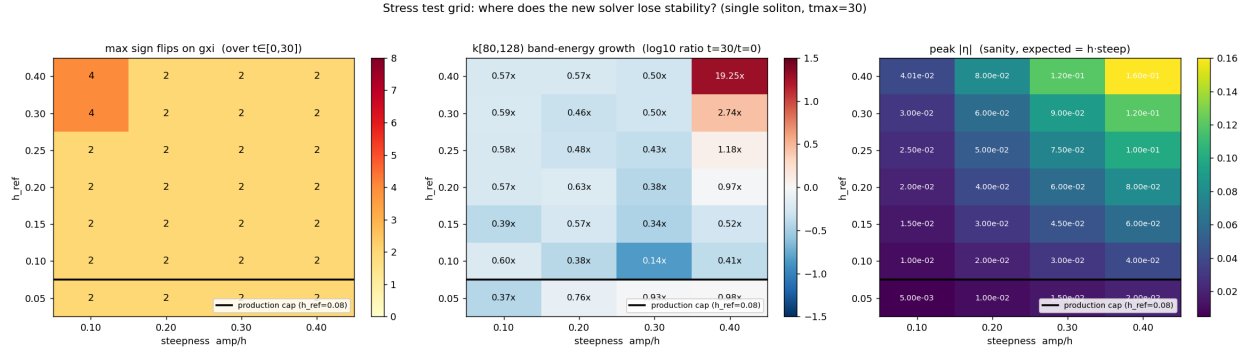


Figure 5: Stress test grid (single soliton, $t_{\max} = 30$). Left: max sign flips on $G\xi$ (misses the failure). Middle: $\log_{10}$ of the $k[80,128]$ growth ratio — the corner ($h_{\text{ref}}=0.40, \alpha=0.40$) glows red. Right: peak $|\eta|$ for sanity, equal to $h_{\text{ref}} \cdot \alpha$ as expected. Black horizontal line marks the production cap $h_{\text{ref}} = 0.08$, which sits well inside the safe region.